

Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • Part2

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Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

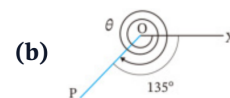
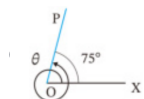
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q001 **Written**

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Classified Practice

Defining Angles and Basic Trigonometric Functions

Q002 **Written****Draw the terminal side for 300° and -450° .**

Q003 Written

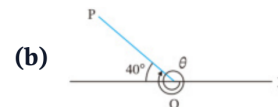
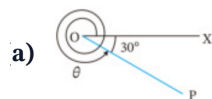
Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q005 **Written****Draw the terminal side for 500° and -120° .**

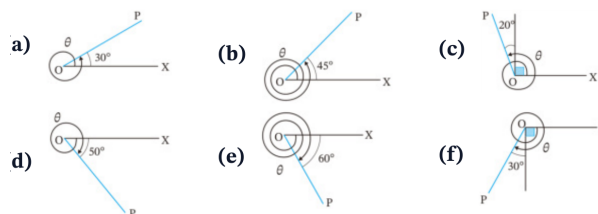
Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).**2** Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Q009 Written**Express** $610^\circ, 920^\circ, 840^\circ, -430^\circ, -960^\circ, -20^\circ$ **as** $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Q011 MCQ

The principal angle coterminal with 765° is:**Choose the correct option.**A 45° B 135° C 225° D 315°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Choose the correct option.**A -50° B 50° C 310° D 410°

Q013 MCQ**The terminal side of 540° lies on the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Choose the correct option.**

A QI

B QII

C QIII

D QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Choose the correct option.**

- A $75^\circ + 360^\circ n$
 - B $285^\circ + 360^\circ n$
 - C $-285^\circ + 360^\circ n$
 - D $360^\circ n - 75^\circ$
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Choose the correct option.**

- A counter-clockwise
- B clockwise
- C radial only
- D undefined

Q017 MCQ**The terminal side of 1080° is the:****Choose the correct option.**

- A positive x -axis
- B negative x -axis
- C positive y -axis
- D negative y -axis

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Choose the correct option.**A $360^\circ n$ B $0^\circ n$ C $180^\circ + 360^\circ n$ D $90^\circ + 360^\circ n$

Q019 MCQ**The principal angle for -990° is:****Choose the correct option.**A 90° B 180° C 270° D -270°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Choose the correct option.

A 25°

B 155°

C 205°

D 335°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert 150° to radians:**Choose the correct option.**

- A $\frac{5\pi}{6}$
B $\frac{3\pi}{5}$
C $\frac{5\pi}{3}$
D $\frac{3\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Choose the correct option.**A -120° B -150° C 150° D -300°

Q030 MCQ

For $r = 5$, $\theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

- A $\frac{5\pi}{3}$
- B $\frac{10\pi}{3}$
- C $\frac{25\pi}{3}$
- D 10π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

- A π
- B 2π
- C 4π
- D 8π

Q032 MCQ

The radian measure of a full turn is:**Choose the correct option.**

- A π
 - B 2π
 - C 180
 - D 360
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

- A 210°
- B 270°
- C 315°
- D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

- A $\frac{\pi}{6}$
- B $-\frac{\pi}{6}$
- C $\frac{5\pi}{6}$
- D $\frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:**Choose the correct option.**

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

Q036 MCQ**The angle $\frac{5\pi}{3}$ lies in:****Choose the correct option.**

A QI

B QII

C QIII

D QIV

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6} \pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4} \pi$

(e) $\cos \frac{5}{3} \pi$

(f) $\cos \frac{7}{6} \pi$

(g) $\tan \frac{2}{3} \pi$

(h) $\tan \frac{5}{4} \pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6} \right)$

(k) $\sin \left(-\frac{5}{4} \pi \right)$

(l) $\sin \left(-\frac{5}{3} \pi \right)$

(m) $\cos \left(-\frac{\pi}{4} \right)$

(n) $\cos \left(-\frac{5}{6} \pi \right)$

(o) $\cos \left(-\frac{4}{3} \pi \right)$

(p) $\tan \left(-\frac{4}{3} \pi \right)$

(q) $\tan \left(-\frac{11}{6} \pi \right)$

(r) $\tan \left(-\frac{3}{2} \pi \right)$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q042 Written

The panel height is $h(t) = 3 \sin\left(\frac{2\pi}{3}t - \frac{\pi}{4}\right) + 1.5$. Find the exact height at $t = 0$.

Q043 MCQ

 $\sin \frac{\pi}{6}$ equals:

Choose the correct option.

A 0

B $\frac{1}{2}$ C $\frac{\sqrt{3}}{2}$

D 1

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:

Choose the correct option.

A -1

B $-\frac{1}{2}$ C $\frac{1}{2}$ D $\frac{\sqrt{3}}{2}$

Q045 MCQ $\tan \frac{3\pi}{4}$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin\left(-\frac{\pi}{2}\right)$ equals:

Choose the correct option.

- A -1
- B 0
- C 1
- D undefined

Q047 MCQ $\tan \frac{\pi}{2}$ is:**Choose the correct option.**

A 0

B 1

C -1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q049 MCQ**If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:****Choose the correct option.**

- A** x
- B** y
- C** x/y
- D** $1/x$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:

Choose the correct option.

- A -1
- B 0
- C 1
- D undefined

Q051 MCQ $\sin \frac{11\pi}{6}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}$, $\cos \frac{4\pi}{3}$, $\tan \frac{7\pi}{4}$.

Q053 Written

Find $\sin(-\frac{5\pi}{6})$, $\cos(-\frac{7\pi}{6})$, $\tan(-\frac{\pi}{3})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q054 **Written**

For $P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 **Written**

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Q060 Written**Solve** $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Choose the correct option.**

- A $\frac{\pi}{4}, \frac{5\pi}{4}$
- B $\frac{3\pi}{4}, \frac{7\pi}{4}$
- C $\frac{\pi}{4}, \frac{7\pi}{4}$
- D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $0, 2\pi$ D π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$

C π

D $\frac{3\pi}{2}$

Q066 MCQ**For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:****Choose the correct option.**

- A $\frac{\pi}{3}, \frac{4\pi}{3}$
B $\frac{2\pi}{3}, \frac{5\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Choose the correct option.**

- A $\frac{\pi}{2}$
- B $\frac{3\pi}{2}$
- C $\frac{2}{\pi}$
- D 0

Q068 MCQ**For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:****Choose the correct option.**

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C π
- D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Choose the correct option.

A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $\frac{\pi}{4}, \frac{5\pi}{4}$ D $\pi, 2\pi$

Q070 Written**Solve** $\cos \theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Q072 Written**Solve** $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.

Q074 Written**Solve** $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 [Concept Quiz · MCQ](#)

The radian measure of 240° is:

Choose the correct option.

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{2\pi}{3}$
B $\frac{4\pi}{3}, \frac{5\pi}{3}$
C $\frac{2\pi}{3}, \frac{4\pi}{3}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for -300° is:

Choose the correct option.

- A $60^\circ + 360^\circ n$
- B $300^\circ + 360^\circ n$
- C $-60^\circ + 360^\circ n$
- D $120^\circ + 360^\circ n$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\cos \theta = -\frac{\sqrt{2}}{2}$ **for** $0 \leq \theta < 2\pi$.

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 **Concept Quiz · Written**

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$; $-\frac{20\pi}{9}$; 330° ; -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -700°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; 0 , π ; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; $\frac{3\pi}{4}$, $\frac{5\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{11\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; $\frac{3\pi}{4}$, $\frac{5\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{11\pi}{6}$, $\frac{5\pi}{6}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q003 **Written**

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q005 **Written**

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q007 **Written**

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ in QIV, find $\sin \theta$ and $\cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q009 **Written**

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ in QIII, $\cos \theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
B $-\frac{1}{2}$
C $\frac{\sqrt{3}}{2}$
D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Choose the correct option.

- A $\frac{7}{24}$
- B $\frac{24}{7}$
- C $-\frac{7}{24}$
- D $\frac{25}{7}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, with θ in QII, evaluate (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6 \theta - \cos^6 \theta$.

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII, find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin \theta \cos \theta$ and $\sin^3 \theta - \cos^3 \theta$.

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Q023 Written

Given $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{5}{16}$
B $\frac{11}{16}$
C $-\frac{11}{16}$
D $\frac{3}{16}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 Written

Prove $\frac{\sin \theta}{1+\cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}.$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; (b) $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q034 Written

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 Written

Prove $\frac{\tan \theta}{1 + \tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Q037 Written

Prove $\frac{\cos \theta}{1+\sin \theta} = \frac{1-\sin \theta}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Choose the correct option.**

- A Change both sides at once
- B Simplify one side
- C Substitute random values
- D Take a square root

Q039 MCQ**The identity used to replace $1 - \sin^2 \theta$ is:****Choose the correct option.**A $\cos^2 \theta$ B $\tan^2 \theta$ C $\sec^2 \theta$ D $\sin^2 \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q040 MCQ

 $\tan \theta$ should be rewritten as:

Choose the correct option.

A $\sin \theta \cos \theta$

B $\sin \theta / \cos \theta$

C $\cos \theta / \sin \theta$

D $1 / \sin \theta$

Q041 MCQ**To add two trig fractions, first find a:****Choose the correct option.**

- A common numerator
- B common denominator
- C new quadrant
- D reference angle

Classified Practice

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Choose the correct option.**

- A LHS=0
- B LHS=RHS
- C RHS=1
- D undefined

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q047 MCQ**If $s + c = 1$, then sc equals:****Choose the correct option.****A** 0**B** $\frac{1}{2}$ **C** $-\frac{1}{2}$ **D** 1

Classified Practice

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{19}{27}$
B $\frac{5}{27}$
C $-\frac{19}{27}$
D $\frac{11}{27}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

- A $\frac{11}{16}$
B $-\frac{11}{16}$
C $\frac{7}{16}$
D $-\frac{7}{16}$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Choose the correct option.

- A $\frac{\sin \theta}{\cos \theta}$
B $\frac{\cos \theta}{\sin \theta}$
C $\sin \theta \cos \theta$
D $\sec \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Choose the correct option.**A $\sin^2 \theta$ B $\tan^2 \theta$ C $\cos^2 \theta$

D 1

Q053 MCQ**When proving an identity, which side should normally be transformed?****Choose the correct option.**

- A Both sides simultaneously
- B One side only
- C Neither side
- D Only the denominator

Classified Practice

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:**Choose the correct option.**

- A A random angle
- B The Pythagorean identity
- C A logarithm
- D A derivative

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Choose the correct option.

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $-\frac{4}{5}$
- D $\frac{3}{4}$

Quiz MCQ 3 Concept Quiz · MCQ**Which identity is always valid where defined?****Choose the correct option.**

A $\tan \theta = \frac{\cos \theta}{\sin \theta}$

B $1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$

C $\sin \theta + \cos \theta = 1$

D $\sin \theta \cos \theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written**Given** $s + c = \frac{2}{3}$, **find** sc **and** $s^3 + c^3$.

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$

Q018 $-\frac{55\sqrt{7}}{256}$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$

Q019 $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$

Q007 $\frac{3}{5}$, $\frac{4}{5}$

Q020 $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$

Q008 $-\frac{5}{13}$, $\frac{12}{13}$

Q021 $-\frac{11}{16}$

Q009 $-\frac{\sqrt{2}}{2}$, 1

Q022 $\frac{\sqrt{11}}{3}$

Q010 B

Q023 $\frac{3}{2}$

Q011 A

Q024 A

Q012 B

Q025 B

Q013 C

Q026 B

Q014 A

Q027 A

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q004 **Written**

Simplify $\sin\left(\frac{3\pi}{2} - \theta\right) \sec(\pi + \theta) - \tan\left(\theta - \frac{\pi}{2}\right) \cot\left(\frac{5\pi}{2} + \theta\right).$

Q005 Written

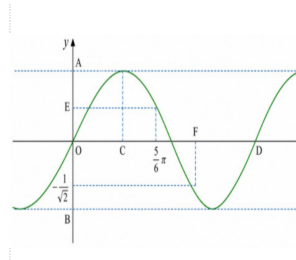
Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, and $y = \tan \theta$, and state each period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .

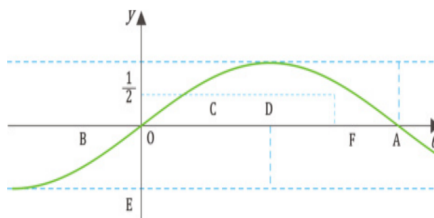


Classified Practice

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .

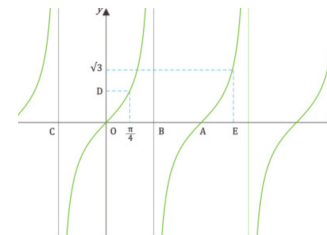
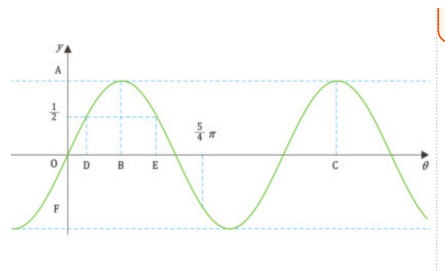
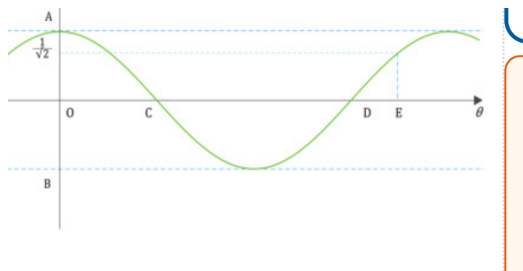


Classified Practice

Properties and Graphs of Trigonometric Functions

Q008 **Written**

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.

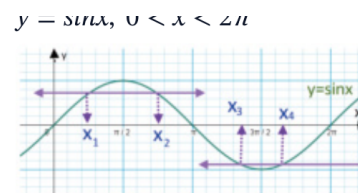


Classified Practice

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q013 **Written****Draw $y = -2|\sin(3x)|$ and find its period.**

Q014 Written

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos \left| x + \frac{\pi}{3} \right|$ and find its period.

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.

Q022 MCQ $\sin(\theta + \pi)$ equals:**Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q024 MCQ $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q026 MCQ **$\sin(-\theta)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q028 MCQ $\sin\left(\frac{3\pi}{2} + \theta\right)$ **equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:

Choose the correct option.

A $\tan \theta$ B $-\tan \theta$ C $\cot \theta$ D $-\cot \theta$

Q030 MCQ**The period of $y = \sin \theta$ is:****Choose the correct option.****A** $\pi/2$ **B** π **C** 2π **D** 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Choose the correct option.**

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D $y > 1$

Q032 MCQ

The period of $y = \tan \theta$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Choose the correct option.**

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Choose the correct option.**

A -1

B 0

C 1

D π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Choose the correct option.**

- A $\pi/4$
- B $\pi/2$
- C π
- D $3\pi/2$

Q036 MCQ**At $\theta = 0$, $\cos \theta$ equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** undefined

Classified Practice

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Choose the correct option.**

- A -1
- B 0
- C 1
- D undefined

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Choose the correct option.**A $[-1,1]$ B $[-3,3]$ C $[0,3]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Q040 MCQ**The period of $y = \cos(\theta/3)$ is:****Choose the correct option.**A $2\pi/3$ B 2π C 3π D 6π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Choose the correct option.**A $\pi/5$ B π C 5π D $2\pi/5$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:

Choose the correct option.

A $[-2, 2]$

B $[-1, 1]$

C $[-1/2, 1/2]$

D $[0, 1/2]$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Choose the correct option.**

- A shifted only
- B reflected and stretched vertically
- C compressed horizontally only
- D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C 3π D 6π

Q046 MCQ**The graph of $y = \sin(\theta - \pi/4)$ is shifted:****Choose the correct option.**

- A left by $\pi/4$
- B right by $\pi/4$
- C up by $\pi/4$
- D down by $\pi/4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Choose the correct option.**A right by $\pi/3$ B left by $\pi/3$ C up by $\pi/3$ D down by $\pi/3$

Q048 MCQ**A horizontal shift changes the sine period from 2π to:****Choose the correct option.****A** π **B** 2π **C** 4π **D** 0

Classified Practice

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Q050 MCQ**The graph of $y = \sin(\theta + \pi/2)$ equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Choose the correct option.**A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Choose the correct option.**

- A $\theta = 0$
- B $\theta = \pi/2$
- C $\theta = \pi$
- D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Choose the correct option.**A $\cos(x)$ B $-\cos(x)$ C $\sin(x)$ D $|\cos(x)|$

Q054 MCQ**In $y = a \sin(b\theta - q)$, the horizontal shift is:****Choose the correct option.**

- A** q
- B** q/b
- C** b/q
- D** a/b

Classified Practice

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C π D 2π

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Choose the correct option.**A $[-1, 1]$ B $[-2, 2]$ C $[-4, 4]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Choose the correct option.**

- A left by $\pi/3$
- B right by $\pi/6$
- C right by $\pi/3$
- D left by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Choose the correct option.**

- A π
 - B 2π
 - C 3π
 - D 4π
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:**Choose the correct option.**

- A vertical scale
- B horizontal shift
- C horizontal scale
- D reflection only

Q060 MCQ**The period of $y = \tan(4\theta + q)$ is:****Choose the correct option.****A** $\pi/4$ **B** $\pi/2$ **C** π **D** 2π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Choose the correct option.**A $[-3,3]$ B $[0,3]$ C $[-1,1]$ D $\mathbb{R}$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos\left(\frac{\pi}{2} - \theta\right)$.

Q064 Written**Simplify** $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan\left(-\frac{3\pi}{4}\right)$ **using an acute reference angle.**

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 Written**Find the period and range of $y = 4 \cos \theta$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.

Q072 Written**Find the period and range of $y = \tan(\theta/4)$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Q074 Written**Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 Written**Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Q078 Written**For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 Concept Quiz · MCQ

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Quiz MCQ 2 Concept Quiz · MCQ

$\sin(\theta + \pi)$ equals:

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 **Concept Quiz · MCQ****The range of $y = 2 \cos \theta$ is:****Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $\mathbb{R}$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

- A left by $\frac{\pi}{4}$
- B right by $\frac{\pi}{4}$
- C up by $\frac{\pi}{4}$
- D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $^{-1}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\max} = 2\pi, T_{\min} = 2\pi, T_{\text{mid}} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ and (b) $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ and (b) $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 **Written**

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ and give the general solution.

Q005 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos \theta > -\sqrt{3}$, (b) $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ **for** $0 \leq \theta < 2\pi$.

Q009 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos^2 \theta - 3 \sin \theta = 0$, (b) $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Q013 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Choose the correct option.**A 2π B π C $\pi/2$ D 4π

Q019 MCQ**The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:****Choose the correct option.****A $\pi/6, 11\pi/6$** **B $\pi/3, 5\pi/3$** **C $2\pi/3, 4\pi/3$** **D $\pi/4, 7\pi/4$**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Choose the correct option.

A $\sin \theta = 1/2$

B $\sin \theta = -2$

C $\sin \theta = 0$

D $\sin \theta = -1/2$

Q021 MCQ

The quadratic challenge factors into roots:**Choose the correct option.**

- A 1 and $\sqrt{3}$
- B 2 and $\sqrt{3}$
- C 2 and 3
- D $\pi/3$ and 2

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Choose the correct option.

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

Q023 MCQ**The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:****Choose the correct option.****A** $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$ **B** $[0, \pi/4]$ **C** $(\pi/2, \pi)$ **D** $[\pi, 2\pi)$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:

Choose the correct option.

- A strictly greater
- B strictly less
- C greater than or equal to
- D never

Q025 MCQ**The 5-14 challenge reduces to which cosine condition?****Choose the correct option.**

A $\cos \theta > 1/2$

B $-1 < \cos \theta < 1/2$

C $\cos \theta \leq -1$

D $\cos \theta \geq 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Choose the correct option.**

A keep it

B reject it

C square it

D divide by 2

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Choose the correct option.**

A $\sin^2 \theta + \cos^2 \theta = 1$

B $\tan \theta = \sin \theta$

C $\cos \theta = 1$

D the tangent period

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:

Choose the correct option.

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

Q029 MCQ**In the rational challenge, which angle must be excluded?****Choose the correct option.****A** 0**B** $\pi/4$ **C** $\pi/2$ **D** π

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Choose the correct option.

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Choose the correct option.

A $\pi/12$

B $\pi/6$

C $\pi/3$

D $\pi/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Choose the correct option.

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

Q033 MCQ**The 5-16 challenge is equivalent to:****Choose the correct option.**

A $|\sin(t)| \geq 1/2$

B $\sin(t) = 0$

C $\cos(t) > 1/2$

D $\tan(t) = 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Choose the correct option.**A $\cos(x)$ B $\sin(x)$ C $\tan^2 x$

D 1

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan \theta = -1$ is:

Choose the correct option.

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Choose the correct option.

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Choose the correct option.

A $\sin \theta = -2$

B $\sin \theta = 1/2$

C $\sin \theta = 2$

D $\sin \theta = -1/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The first operation in a shifted-argument problem is to:

Choose the correct option.

- A divide by $\cos \theta$
- B set t equal to the whole argument and find its range
- C change radians to degrees
- D square both sides

Quiz WRITTEN 5 Concept Quiz · Written

Solve $2 \cos \theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.

Quiz WRITTEN 7 Concept Quiz · Written

Solve $2 \sin^2 \theta + \sin \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B
- Q002** (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B
- Q003** (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B
- Q004** $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A
- Q005** (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C
- Q006** (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$. **Q025** B
- Q007** (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B
- Q008** $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q027** A
- Q009** (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q028** A
- Q010** (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q029** C
- Q011** (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q018** B
- Q019** B

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Classified Practice

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Classified Practice

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Classified Practice

Advanced Theorems and Applications

Q004 **Written****Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.**

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ **in QII** and $\cos \beta = -\frac{3}{5}$ **in QIII**, **find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$.

Classified Practice

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled* α, β , and *the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Q007 Written**Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.**

Classified Practice

Advanced Theorems and Applications

Q008 Written

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Classified Practice

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Classified Practice

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ with α in QII, find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Classified Practice

Advanced Theorems and Applications

Q012 **Written**

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 Written

Given $\cos \alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice

Advanced Theorems and Applications

Q014 **Written**

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi,$
 $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi.$

Classified Practice

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q017 Written**Solve the given equation** $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q018 **Written**

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Classified Practice

Advanced Theorems and Applications

Q020 Written

Rewrite the **given** expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in **harmonic-addition form**.

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Classified Practice

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.

Q023 Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q024 **Written**

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Classified Practice

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:

Choose the correct option.

A $t = \sin \theta$

B $t = \cos \theta$

C $t = \tan \theta$

D $t = \theta$

Q029 MCQ**The maximum of $\sin^2 \theta + \cos \theta + 1$ is:****Choose the correct option.****A** 0**B** 1**C** 9/4**D** 4

Classified Practice

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Choose the correct option.**A $[-2,2]$ B $[-1,1]$ C $[0,1]$

D all real numbers

Q031 MCQ

The identity for $\sin(a+b)$ is:**Choose the correct option.**

- A $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- B $\sin \alpha \sin \beta$
- C $\cos \alpha \cos \beta$
- D $\tan \alpha + \tan \beta$

Classified Practice

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Choose the correct option.**A $(\sqrt{6}-\sqrt{2})/4$ B $(\sqrt{6}+\sqrt{2})/4$ C $1/2$ D $\sqrt{3}/2$

Q033 MCQ**A missing trig value is selected using:****Choose the correct option.**

- A the quadrant
- B the denominator
- C the period only
- D a graph scale

Classified Practice

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Choose the correct option.**

A k

B m

C $1/m$ D $m+k$

Q035 MCQ**The angle formula uses denominator:****Choose the correct option.****A** $\frac{1+m_1m_2}{1-m_1m_2}$ **B** $\frac{m_1-m_2}{m_1+m_2}$ **C** $\frac{m_1+m_2}{m_1-m_2}$ **D** $\frac{m_1m_2}{1-m_1m_2}$

Classified Practice

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Choose the correct option.

A θ_1

B $\pi - \theta_1$

C $2\theta_1$

D $-\theta_1$ only

Q037 MCQ **$\sin(2a)$ equals:****Choose the correct option.**

- A $\sin \alpha$
- B $2 \sin \alpha \cos \alpha$
- C $\cos^2 \alpha$
- D $\tan \alpha$

Classified Practice

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2\alpha)$ can be written as:**Choose the correct option.**A $\cos^2 \alpha - \sin^2 \alpha$ B $2\sin \alpha$ C $\tan \alpha$ D $1 + \sin \alpha$

Q039 MCQ**For theta in QIV, sin theta is:****Choose the correct option.**

- A positive
- B negative
- C zero always
- D undefined

Classified Practice

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Choose the correct option.**A $(1 - \cos \alpha)/2$ B $(1 + \cos \alpha)/2$ C $\tan \alpha$ D $\cos 2\alpha$

Q041 MCQ**The sign of a half-angle root is chosen from:****Choose the correct option.**

- A the original quadrant interval
- B the coefficient only
- C the denominator
- D the period of $\tan$

Classified Practice

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ**For double-angle equations, first rewrite in:****Choose the correct option.**

- A one trig function
- B two unrelated angles
- C degrees only
- D a logarithm

Classified Practice

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Choose the correct option.**A $\sin\theta(2\cos\theta + 1)$ B $\cos\theta(2\sin\theta - 1)$ C $\sin\theta + \cos\theta$ D $2\sin\theta$

Q045 MCQ**A tangent asymptote is always:****Choose the correct option.**

- A included
- B excluded
- C a maximum
- D an endpoint

Classified Practice

Advanced Theorems and Applications

Q046 MCQ

For $a \sin \theta + b \cos \theta$, r is:**Choose the correct option.**A $a+b$ B $\sqrt{a^2+b^2}$ C $a-b$ D ab

Q047 MCQ**The harmonic target form is:****Choose the correct option.****A** $r \sin(\theta + \alpha)$ **B** $r \cos \theta$ **C** $a \tan \theta$ **D** $\sin \theta$ only

Classified Practice

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Choose the correct option.**

- A $P(a,b)$
- B $P(r,\theta)$
- C the origin only
- D the x intercept

Q049 MCQ**After synthesis, t is usually:****Choose the correct option.****A** $\theta + \alpha$ **B** θ^2 **C** $\tan \theta$ **D** $\cos \theta$

Classified Practice

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < \frac{3}{2}$ is:

Choose the correct option.

- A never true
- B always true
- C true only at π
- D undefined

Q051 MCQ**In a strict inequality, equality endpoints are:****Choose the correct option.**

- A included
- B excluded
- C doubled
- D random

Classified Practice

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $[-3,3]$

Q053 MCQ**For a double-angle maximum problem, use:****Choose the correct option.**

- A the double-angle formula then a bounded variable
- B a logarithm
- C only a graph
- D no substitution

Classified Practice

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Choose the correct option.**

- A an allowed extremum
- B rejected
- C always the maximum
- D always the minimum

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 2 Concept Quiz · MCQ

The addition-theorem value $\sin 75^\circ$ is:

Choose the correct option.

A $(\sqrt{6} + \sqrt{2})/4$

B $(\sqrt{6} - \sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 Concept Quiz · MCQ

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{3}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a\theta = \frac{\pi}{2}, \pi$; $b\ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}, \min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Classified Practice

Laws of Exponents and Roots

Q001 **Written****Find:** (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.

Classified Practice

Laws of Exponents and Roots

Q003 Written

Find: (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q004 Written

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q005 Written

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Classified Practice

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3a^{-4} \div (2a)^{-2}$.

Classified Practice

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2).$

Q009 Written**Find** $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Classified Practice

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[3]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Q011 Written**Find** $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Classified Practice

Laws of Exponents and Roots

Q014 **Written**

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Classified Practice

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Classified Practice

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Classified Practice

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

Classified Practice

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{1/3}]^{1/2}$, $[(1/8)^{1/2}]^{1/3}$, $[(1/16)^{1/4}]^{1/2}$

Classified Practice

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

Classified Practice

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7^6}\sqrt{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ **and** $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Classified Practice

Laws of Exponents and Roots

Q026 Written

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six **given** expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[5]{a}$ and mixed a,b radicals.

Classified Practice

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Q029 Written

Assess the statement $\sqrt{a^2} = a$ for all real a , using positive and negative examples.

Classified Practice

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Choose the correct option.**

A 8

B $1/8$ C -8 D $1/6$

Q031 MCQ**When multiplying equal bases, we:****Choose the correct option.**

- A subtract exponents
- B add exponents
- C multiply bases only
- D take reciprocals

Classified Practice

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 MCQ**A negative exponent means:****Choose the correct option.**

- A multiply by -1
- B take the reciprocal power
- C set the base to zero
- D add one

Classified Practice

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Choose the correct option.**

- A 25
- B 400
- C 625
- D 1600

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Choose the correct option.**

A -2

B -1

C 2

D 6

Classified Practice

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Choose the correct option.**

A a^7b^6

B $1/(a^7b^6)$

C $a^{-7}b^{-6}$

D $1/(a^6b^7)$

Q037 MCQ**The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:****Choose the correct option.****A** $1/3$ **B** 1**C** 3**D** 9

Classified Practice

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:

Choose the correct option.

A 2

B 4

C 8

D 16

Q039 MCQ $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Choose the correct option.**

A 4

B 5

C 6

D 9

Classified Practice

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Choose the correct option.**A $1/2$ B $1/4$

C 2

D 4

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

- A $\sqrt[3]{2}$
 - B $2\sqrt[3]{2}$
 - C $3\sqrt[3]{2}$
 - D 0
-
-
-

Classified Practice

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 MCQ**When combining roots with the same index, multiply:****Choose the correct option.**

- A the indices
- B the radicands
- C the exponents only
- D the signs only

Classified Practice

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Choose the correct option.**

A 0

B 1

C 2

D 4

Q045 MCQ **$\sqrt[5]{2^4}$ in rational-index form is:****Choose the correct option.**

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

Classified Practice

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Choose the correct option.**

A 3

B 6

C 9

D 18

Q047 MCQ**A reciprocal square root is written as:****Choose the correct option.**

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^2

Classified Practice

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Choose the correct option.**

A 27

B $1/9$ C $1/27$ D $1/81$

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

- A $3/4$
- B $4/3$
- C $-3/4$
- D 3

Classified Practice

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

Choose the correct option.

A $4/9$

B $9/4$

C $16/9$

D $27/64$

Q051 MCQ**The 6^{-3} challenge simplifies to:****Choose the correct option.**A $3/10$ B $10/3$ C $100/9$ D $9/100$

Classified Practice

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3} 2^{1/2} \div 2^{7/6}$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Q053 MCQ $4^{5/6} 2^{4/3}$ equals:**Choose the correct option.****A** 4**B** 6**C** 8**D** 16

Classified Practice

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

Choose the correct option.

A $a^{1/3}$ B $a^{2/3}$ C $a^{5/3}$

D a

Q055 MCQ

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:

Choose the correct option.

A 2

B 3

C 6

D 12

Classified Practice

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

- A integer powers
- B combining rational exponents
- C logarithms
- D trigonometry

Q057 MCQ**The printed decimal in the 6-4 challenge is:****Choose the correct option.**

- A 0.04
 - B 0.004
 - C 0.4
 - D 4
-
-
-

Classified Practice

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:**Choose the correct option.**

- A a always
- B $-a$ always
- C $|a|$
- D 0

Quiz MCQ 1 **Concept Quiz · MCQ**

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

- A same-base multiplication
- B power of a power
- C reciprocal law
- D root product

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

- A 2
- B 5
- C 10
- D 125

Quiz MCQ 3 Concept Quiz · MCQ

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512

B $1/64$

C $1/512$

D 64

Classified Practice

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Choose the correct option.

A a

B $-a$

C 0

D a^2

Quiz WRITTEN 5 Concept Quiz · Written**Simplify** $a^{-3}a^7 \div a^2$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 Concept Quiz · Written**Evaluate** $125^{-2/3}$.

Classified Practice

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

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Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Classified Practice

Exponential Functions and Their Applications

Q001 **Written**

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Q002 Written**Draw $y = 2^x$ and $y = (1/2)^x$.**

Classified Practice

Exponential Functions and Their Applications

Q003 Written

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Q004 Written**Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.**

Classified Practice

Exponential Functions and Their Applications

Q005 **Written**

Order the given sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Q006 Written

Order the three displayed Try sets: $5^0, 5^{-4}, 5^{3/2}$; $(1/3)^7, (1/3)^2, (1/3)^{-5}$; **and** $3^0, 3^2, 3^{-2}$.

Classified Practice

Exponential Functions and Their Applications

Q007 Written

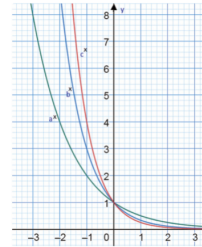
Order the given sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Classified Practice

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c .



Classified Practice

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

Classified Practice

Exponential Functions and Their Applications

Q011 **Written**

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Classified Practice

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Classified Practice

Exponential Functions and Their Applications

Q015 **Written**

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Classified Practice

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Classified Practice

Exponential Functions and Their Applications

Q019 Written

Solve the three given extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

Q020 Written**For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.**

Classified Practice

Exponential Functions and Their Applications

Q021 Written

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Q022 MCQ

Which base gives a strictly decreasing exponential graph?**Choose the correct option.**

A 2

B 3

C $1/2$ D π

Classified Practice

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

Q024 MCQ

The range of $y = 5^x$ is:**Choose the correct option.**A $\mathbb{R}$ B $[0, \infty)$ C $(0, \infty)$ D $(-\infty, 0)$

Classified Practice

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?

Choose the correct option.

A $(0, \pi)$

B $(1, \pi)$

C $(2, \pi)$

D $(-1, \pi)$

Q026 MCQ**For $0 < a < 1$ and $r < s$, which is true?****Choose the correct option.**

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

Classified Practice

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Choose the correct option.**

- A add exponents
- B make the bases equal
- C take logarithms immediately
- D ignore the base

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Choose the correct option.**A 3^{-2} B 3^0 C 3^2

D They are equal

Classified Practice

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Choose the correct option.

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

Q030 MCQ**Solving $4^{x-1} = 8^{x+3}$ gives:****Choose the correct option.**

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

Classified Practice

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Choose the correct option.**

- A the same exponent order
- B a reversed exponent order
- C no base conversion
- D squaring both sides

Q032 MCQ

The solution of $3^x = 81$ is:**Choose the correct option.**

A 2

B 3

C 4

D 5

Classified Practice

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Choose the correct option.

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

Classified Practice

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Choose the correct option.**

A $(3 - \sqrt{73})/2$

B $(3 + \sqrt{73})/2$

C 8

D 16

Q036 MCQ

The challenge quadratic in t is:**Choose the correct option.**

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

Classified Practice

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Choose the correct option.**

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

Q038 MCQ**For extrema, the first required range is the range of:****Choose the correct option.****A** x **B** $t = a^x$ **C** a **D** y only

Classified Practice

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Choose the correct option.**

A -8

B -4

C 0

D 4

Q040 MCQ**The C06-K02 challenge has minimum value:****Choose the correct option.****A** 9 at $x = 0$ **B** 9 at $x = 2$ **C** 27 at $x = 0$ **D** 27 at $x = 2$

Classified Practice

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Choose the correct option.**

A 0

B 1

C 2

D It has no minimum

Q042 MCQ**At which value of x does $a^x + a^{-x}$ attain its minimum?****Choose the correct option.****A** $x = -1$ **B** $x = 0$ **C** $x = 1$ **D** It has no stationary point

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Choose the correct option.

- A range $(0, \infty)$
- B range all reals
- C vertical asymptote
- D decreasing for every a

Quiz MCQ 2 Concept Quiz · MCQ

For base 0.5, increasing the exponent makes the value:

Choose the correct option.

- A larger
- B smaller
- C unchanged
- D zero

Classified Practice

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The positive-substitution condition is:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D t can be any real

Quiz MCQ 4 Concept Quiz · MCQ

For $y = 2(t - 1)^2 + 9$, the minimum is:

Choose the correct option.

A 0

B 1

C 9

D 11

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 5 Concept Quiz · Written

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Quiz WRITTEN 6 Concept Quiz · Written**Order** 2^{-3} , 2^0 , $2^{5/2}$.

Classified Practice

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

Quiz WRITTEN 8 Concept Quiz · Written

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, 3/2: strictly increasing. Bases 2/3, 1/4, 1/10: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Classified Practice

Introduction to Logarithms and Their Properties

Q001 **Written**

Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Q002 Written

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Classified Practice

Introduction to Logarithms and Their Properties

Q003 **Written**

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Q004 Written

If $\log_5 25 = x$, **evaluate** $\log_{1/2} x$ **without a calculator**.

Classified Practice

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1$, $\log_5 5$, $\log_2 8$, $\log_2 16$, $\log_5(1/5)$, $\log_7(49^{3/2})$.

Q006 Written

Find the six Try values $\log_5 1$, $\log_3 3$, $\log_4 4$, $\log_3 27$, $\log_2(1/4)$, $\log_5(1/125)$.

Classified Practice

Introduction to Logarithms and Their Properties

Q007 **Written**

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

Q008 Written**Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.**

Classified Practice

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4 \log_5 3 - 2 \log_5 15 - \log_5 45$.

Classified Practice

Introduction to Logarithms and Their Properties

Q011 Written

Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.

Classified Practice

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24$, $\log_{10}(27/4)$, $\log_{10} 45$, $\log_{10}(24/75)$ in terms of a, b .

Q014 Written

Given $\log_{10} 2 = a$, $\log_{10} 3 = b$, express the four Try arguments 72 , $3^8/10$, 15 , 12 in terms of a, b .

Classified Practice

Introduction to Logarithms and Their Properties

Q015 **Written**

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Q016 Written

If $\log_c 3 = x$, $\log_c 2 = y$, **write** $\log_c 12$, $\log_c(3/8)$, $\log_c \sqrt{6}$, $\log_2 3$, $\log_{3^b}(2^b c)$ **in terms of** x, y .

Classified Practice

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Classified Practice

Introduction to Logarithms and Their Properties

Q019 **Written**

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Q020 Written**Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.**

Classified Practice

Introduction to Logarithms and Their Properties

Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Q022 Written**Solve the four Try products of logarithms after making the bases the same.**

Classified Practice

Introduction to Logarithms and Their Properties

Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

Q024 Written**Simplify** $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ **for** $x > 1$.

Classified Practice

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

Classified Practice

Introduction to Logarithms and Their Properties

Q027 Written

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.

Q028 Written**Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.**

Classified Practice

Introduction to Logarithms and Their Properties

Q029 Written

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Q030 Written

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2 \log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Classified Practice**Introduction to Logarithms and Their Properties****Q031** Written**Order the four Try sets of logarithms using monotonicity and change of base.**

Q032 Written**Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.**

Classified Practice

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Q034 MCQ $\log_2 32$ equals:**Choose the correct option.****A** 2**B** 4**C** 5**D** 32

Classified Practice

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:

Choose the correct option.

A -2

B -1

C 1

D 2

Q036 MCQ $\log_5(1/125)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:

Choose the correct option.

A 1

B 2

C $3/2$ D $7/2$

Q038 MCQ

$\log_a M + \log_a N$ equals:

Choose the correct option.

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

Classified Practice

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Choose the correct option.**

A $\log_5(2 + 3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

Q040 MCQ**If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:****Choose the correct option.**

- A** $a + b$
- B** $2a + 3b$
- C** $3a + b$
- D** $3a + 2b$

Classified Practice

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Choose the correct option.

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

Q042 MCQ $\log_{16} 64$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Choose the correct option.**

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

Q044 MCQ $\log_4 16 \cdot \log_9 3$ equals:

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Classified Practice

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Choose the correct option.**

A 6

B 8

C 12

D 24

Q046 MCQ

The domain of $y = \log_a x$ is:**Choose the correct option.**

A all real x

B $x \geq 0$ C $x > 0$ D $x \neq 0$

Classified Practice

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Choose the correct option.

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

Q048 MCQ**For $a > 1$ and $M < N$, which is true?****Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal**D** Cannot compare

Classified Practice

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Choose the correct option.**

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

Classified Practice

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Choose the correct option.

A $\log_a M^{-2}$

B $\log_a(-2M)$

C $\log_a(M - 2)$

D $2 \log_a M$

Q052 MCQ

The base of $\log_a M$ is:**Choose the correct option.**A M B a C p D $1/a$

Classified Practice

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Choose the correct option.**A a B M C p D 0

Q054 MCQ $\log_2(1/8)$ equals:**Choose the correct option.****A** -3**B** -2**C** 2**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:

Choose the correct option.

A 0

B 1

C 2

D 4

Q056 MCQ $\log_3 1$ **equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** 3

Classified Practice

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:

Choose the correct option.

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

Q058 MCQ **$2 \log_3 5$ equals:****Choose the correct option.****A** $\log_3 10$ **B** $\log_3 25$ **C** $\log_3 7$ **D** $\log_3(5/2)$

Classified Practice

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Choose the correct option.

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Choose the correct option.

A a

B $1 - a$

C $1 + a$

D $-a$

Classified Practice

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:

Choose the correct option.

A $2/3$ B $3/2$ C $5/3$

D 3

Q062 MCQ $\log_{25} 125$ equals:

Choose the correct option.

A $1/2$ B $3/2$

C 2

D $5/2$

Classified Practice

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:

Choose the correct option.

A 0

B 1

C 2

D $\log_a b$

Q064 MCQ

$\log_2 8 \cdot \log_8 2$ equals:

Choose the correct option.

A $1/3$

B 1

C 3

D 8

Classified Practice

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Choose the correct option.**

A 2

B 4

C 8

D 10

Q066 MCQ

The range of $y = \log_a x$ is:**Choose the correct option.**A $x > 0$ B $\mathbb{R}$ C $[0, \infty)$ D $y > 0$

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Choose the correct option.

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

Quiz MCQ 2 Concept Quiz · MCQ

$\log_2 8 + \log_2 4$ equals:

Choose the correct option.

A 2

B 3

C 5

D 12

Classified Practice

Introduction to Logarithms and Their Properties

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ equals:

Choose the correct option.

A $1/2$

B $3/2$

C 2

D 4

Quiz MCQ 4 Concept Quiz · MCQ

For $a > 1$, $M < N$ implies:

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written

Find $\log_3(1/27)$.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\log_5 25 + \log_5 5$.

Classified Practice

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Quiz WRITTEN 8 Concept Quiz · Written

State the domain and range of $y = \log_a x$.

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4(1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1 A****Quiz MCQ 2 C****Quiz MCQ 3 B****Quiz MCQ 4 B****Quiz WRITTEN 5 -3** **Quiz WRITTEN 6 3****Quiz WRITTEN 7 $3/2$** **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Classified Practice

Applications of Logarithms

Q001 **Written****Solve** $\log_2 x = 4$.

Q002 Written**Solve** $\log_2(2x - 1) = \log_2 3$.

Classified Practice

Applications of Logarithms

Q003 **Written**

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Q004 Written**Solve** $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ **completely.**

Classified Practice

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.

Classified Practice

Applications of Logarithms

Q007 **Written**

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 Written

Solve the **given** challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Classified Practice

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16 \log_2 x + 3 < 0$.

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

Classified Practice

Applications of Logarithms

Q011 **Written****Solve the given groups: three equations, three inequalities, and three exact logarithmic values.**

Q012 Written**Solve** $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x .

Classified Practice

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Classified Practice

Applications of Logarithms

Q015 **Written**

Solve the three given extrema problems, each with a different base and interval.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Classified Practice

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 Written**Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.**

Classified Practice

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Q020 Written**Solve** $\log_x(x + 2) \leq \log_x(4 - x)$ **for all real** x .

Classified Practice

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Choose the correct option.**A $x < 5$ B $x \geq 5$ C $x > 5$

D all real x

Q022 MCQ**If $\log_2(2x - 1) = \log_2 3$, then $x =$:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Choose the correct option.**

A 1

B 2

C 4

D 8

Q024 MCQ**For same-base logarithms, equality of logs gives:****Choose the correct option.**

- A arguments equal
- B bases equal
- C arguments add
- D no result

Classified Practice

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A $M > N$

B $M < N$

C $M = N$

D $MN < 0$

Q026 MCQ**For base $0 < a < 1$, the inequality direction is:****Choose the correct option.**

- A preserved
- B reversed
- C removed
- D squared

Classified Practice

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Choose the correct option.**A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

Q028 MCQ

The warm-up intersection is:**Choose the correct option.**A $0 < x < 5$ B $5 < x < 6$ C $x > 6$ D $-3 < x < 6$

Classified Practice

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Choose the correct option.**

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ**The inequality $5t^2 - 16t + 3 < 0$ gives:****Choose the correct option.****A** $t < 1/5$ **B** $1/5 < t < 3$ **C** $t > 3$ **D** $0 < t < 1$

Classified Practice

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Choose the correct option.**A $t > 0$ B $t \in \mathbb{R}$ C $t < 0$ D $t = 1$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

Choose the correct option.

A $x < 1/9$

B $1/9 < x < 27$

C $x > 27$

D $0 < x < 3$

Classified Practice

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$

B $t = 1$

C $t = 2$

D $t = 3$

Q034 MCQ

The correct first step for extrema is:**Choose the correct option.**

- A differentiate in x
- B set $t = \log_a x$
- C ignore the interval
- D take $x = 0$

Classified Practice

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Choose the correct option.**A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

Q036 MCQ**The water-tank challenge maximum health rating is:****Choose the correct option.****A** 3**B** 4**C** 5**D** 8

Classified Practice

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Choose the correct option.**

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

Q038 MCQ**If $-9 < \log_{10} N < -8$, the first non-zero digit is at:****Choose the correct option.**

- A 8th
- B 9th
- C 10th
- D 11th

Classified Practice

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Choose the correct option.**

A 2

B e

C 10

D any base

Q040 MCQ**The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:****Choose the correct option.****A** (0,1)**B** (1,4)**C** (0,4)**D** empty

Classified Practice

Applications of Logarithms

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $\log_5(x - 1) = 2$.

Choose the correct option.

A $x = 26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 Concept Quiz · MCQ

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.**Choose the correct option.**A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$.

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

Classified Practice

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.

Classified Practice

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 Concept Quiz · Written

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

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Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Q002 Written**Find the average rate of change for the three given Try functions on their stated intervals.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q003 **Written**

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q005 Written

Find the **given** limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 Written**Find the three given Try limits, including polynomial and h-quotient forms.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q007 **Written****Find all nine given Exercise limits, including x-limits and h-limits.**

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 Written**Differentiate the three given Try functions using the definition.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine given Exercise functions using the definition.**

Q016 Written

A regular hexagon has side length x . Write its area and differentiate it using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Q018 MCQ**The average rate of change from a to b is:****Choose the correct option.**

A $[f(a)+f(b)]/(b-a)$

B $[f(b)-f(a)]/(b-a)$

C $f'(a)$

D $b-a$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ**The average area-rate for diameter 10 to 18 is:****Choose the correct option.**A 7π B 14π C 28π D 56π

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Choose the correct option.**

- A setting $h = 1$
- B cancelling the common h
- C differentiating first
- D squaring h

Q022 MCQ**A direct-substitution limit is valid when:****Choose the correct option.**

- A the expression is defined
- B the denominator is zero
- C the numerator is infinite
- D h is negative

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:**Choose the correct option.**

- A factor/cancel first
- B differentiate twice
- C ignore the denominator
- D change the base

Q024 MCQ

In the finite-limit challenge, k equals:**Choose the correct option.**

A 3

B 5

C 6

D 9

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Choose the correct option.**

- A its value at the point
- B zero
- C infinity
- D the slope

Q026 MCQ

The definition of the differential coefficient at a is:**Choose the correct option.**

A $\lim [f(a+h)-f(a)]/h$

B $f(a+h)+f(a)$

C $[f(a)-f(0)]/a$

D $h/f(a)$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Choose the correct option.**

A -12

B -6

C 6

D 12

Q028 MCQ**The stone's velocity at 10 s is:****Choose the correct option.**

- A 49
 - B 98
 - C 196
 - D 4.9
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Choose the correct option.**A $h \rightarrow 0$ B $h \rightarrow 1$ C $x \rightarrow 0$ D $a \rightarrow \infty$

Q030 MCQ**The derivative function is:****Choose the correct option.**

- A the coefficient at one point
- B the differential coefficient for every x
- C the average rate only
- D the original function

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Choose the correct option.**A $x+3$ B $2x+3$ C $2x$ D x^2

Q032 MCQ

The area of a regular hexagon of side x is:**Choose the correct option.**

- A $3\sqrt{3}x$
 - B $(3\sqrt{3}/2)x^2$
 - C $6x^2$
 - D $\sqrt{3}x$
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Choose the correct option.**

- A exist
- B be infinite only
- C equal zero only
- D fail

Q034 MCQ

The derivative of a constant function is:**Choose the correct option.**

A 0

B 1

C x

D the constant

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4

B 1

C 2

D 3

Quiz MCQ 2 Concept Quiz · MCQ**Find** $\lim_{x \rightarrow 2}(x^2 + 1)$.**Choose the correct option.****A** 0**B** 5**C** 2**D** 3

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Choose the correct option.

A 0

B 1

C $2x$

D 3

Quiz MCQ 4 Concept Quiz · MCQ

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Choose the correct option.

A 0

B 1

C 2

D 49

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 **Concept Quiz · Written****Evaluate** $\lim_{h \rightarrow 0} (6 + 2h)$.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

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Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Classified Practice

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Classified Practice

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2, f'(2) = 2, f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.

Classified Practice

Differentiation and Tangent Lines

Q008 Written

Find the quadratic $f(x)$ satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Q009 Written**Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.**

Classified Practice

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

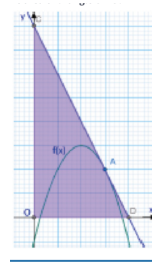
Q011 Written**Find the tangent line for all nine given Exercise functions at their given points.**

Classified Practice

Differentiation and Tangent Lines

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice

Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from $(C(0,2))$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Classified Practice

Differentiation and Tangent Lines

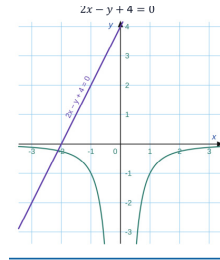
Q015 **Written****Solve all six given** Exercise demands involving external points or prescribed gradients.

Classified Practice

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Choose the correct option.**

A x^{n-1}

B nx^{n-1}

C $n + x$

D x^n

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Choose the correct option.

A πh

B $2\pi r h$

C $2\pi r$

D πr^2

Classified Practice

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:**Choose the correct option.**

A -6

B -3

C 0

D 3

Q020 MCQ**The challenge function can be written as:****Choose the correct option.**

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

Classified Practice

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 MCQ

The condition $f'(0) = -2$ gives:**Choose the correct option.**

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Choose the correct option.**

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ**The challenge coefficient a is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 6

Classified Practice

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Choose the correct option.**A $f(a)$ B a C $f'(a)$ D $1/f'(a)$

Q026 MCQ**The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:****Choose the correct option.**

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

Classified Practice

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Choose the correct option.**

A 2

B 3

C 4

D 8

Q028 MCQ**The area of triangle (ODC) in the challenge is:****Choose the correct option.****A** 8**B** 12**C** 16**D** 32

Classified Practice

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Choose the correct option.**A $((0,0))$ B $((a,f(a)))$ C $((f(a),a))$ D $((1,f'(a)))$

Q030 MCQ**The gradient of $2x - y + 4 = 0$ is:****Choose the correct option.****A** -2**B** -1**C** 1**D** 2

Classified Practice

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:

Choose the correct option.

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

Q032 MCQ**The parallel tangent in the challenge is:****Choose the correct option.**

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the tangent to $y = x^2$ at $((2,4))$.

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 Concept Quiz · MCQ

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

- A three function values
- B two derivatives and one function value
- C one derivative only
- D the tangent gradient only

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 7 **Concept Quiz · MCQ****The tangent equation in point-slope form is:****Choose the correct option.**

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 Concept Quiz · MCQ

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B $1/3$

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q008 $f(x) = 3x^2 - 18x + 27$

Q019 C

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q009 $y = 2x - 8$

Q020 A

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q021 B

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 5$

Q022 B

Q012 16 square units

Q023 A

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q024 C

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q025 C

Q004 $\frac{10}{3}$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q026 B

Q005 $f(x) = x^2 - 2x + 3$

Q016 $y = 2x - 3$

Q027 C

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q017 B

Q028 C

Q018 B

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Classified Practice

Applications of Differentiation: Polynomial Functions

Q001 **Written**

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 Written**Find the extreme values and sketch the three given Try cubics.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q003 **Written**

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 Written**Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q007 **Written**

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b , and the local minimum.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q011 **Written****Solve all six given Exercise problems asking for cubic coefficients from extrema.**

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

Classified Practice

Applications of Differentiation: Polynomial Functions

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q015 **Written**

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q017 **Written**

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Q018 Written

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q019 **Written****Find the parameter ranges in all six given Exercise problems for monotonic increase.**

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k -range for monotonic increase on $x \in [1, \infty)$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 Written**Find the number of distinct real roots for the three given Try equations.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q023 **Written****Find the number of distinct real roots for all nine given cubic equations.**

Q024 Written**Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Q026 Written**Solve the two given Try parameter problems for three or two distinct real roots.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q027 **Written****Solve all four given Exercise parameter problems for the required number of distinct roots.**

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q029 **Written**

Prove the given inequalities using an auxiliary cubic and a derivative sign table.

Q030 Written**Prove all three given cubic inequalities on their stated domains.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 Written**Find the extreme values and sketch the two given quartic functions.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q033 **Written****Find the extreme values and graphs for all six given quartic functions.**

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q035 MCQ

At a local maximum, the sign of f' changes:**Choose the correct option.**A $- \rightarrow +$ B $+ \rightarrow -$ C $+ \rightarrow +$ D $- \rightarrow -$

Q036 MCQ**For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:****Choose the correct option.****A** $a - 1$ **B** a **C** $a + 1$ **D** $1 - a$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q037 MCQ

Absolute extrema on a closed interval require comparing:**Choose the correct option.**

- A stationary points only
- B endpoints only
- C stationary points and endpoints
- D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

Choose the correct option.

A $4\sqrt{2}$

B 9

C 6

D 0

Classified Practice

Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 MCQ**The derivative roots in the challenge are:****Choose the correct option.**A $-4, 2$ B $-2, 4$ C $2, 4$ D $-3, 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Choose the correct option.**A $D > 0$ B $D = 1$ C $D \leq 0$ D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:**Choose the correct option.**

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

- A negative everywhere
- B non-negative everywhere
- C zero everywhere
- D linear

Q044 MCQ

The challenge domain in 7-13 is:**Choose the correct option.**A $(-\infty, \infty)$ B $[1, \infty)$ C $[-1, 1]$ D $(0, 1)$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q045 MCQ

Three distinct roots occur when the x-axis lies:**Choose the correct option.**

- A above both extrema
- B between the local maximum and minimum
- C at an endpoint only
- D at the local maximum only

Q046 MCQ**The 7-14 challenge has how many distinct real roots?****Choose the correct option.****A** 0**B** 1**C** 2**D** 3

Classified Practice

Applications of Differentiation: Polynomial Functions

Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Choose the correct option.**

- A strictly between the extrema
- B above the maximum
- C below the minimum
- D at an extremum

Q048 MCQ

The 7-15 warm-up range is:**Choose the correct option.**

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q049 MCQ

To prove $LHS \geq RHS$, define:**Choose the correct option.**

A $f = LHS + RHS$

B $f = LHS - RHS$

C $f = RHS - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:**Choose the correct option.**

A $2 - 4\sqrt{2}$

B $4\sqrt{2} - 2$

C $2 + 4\sqrt{2}$

D $-2 - 4\sqrt{2}$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q051 MCQ

A repeated root of a derivative may be:**Choose the correct option.**

- A always a local maximum
- B always a local minimum
- C stationary but not an extreme
- D impossible

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Choose the correct option.**A $k = 27$ B $k < 27$ C $k > 27$ D $k \leq 0$ only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q053 MCQ

A local minimum is identified by the change:**Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Q054 MCQ**For $y = x^3 - 6x^2 + 16$, the critical x-values are:****Choose the correct option.****A** 0, 4**B** 1, 3**C** -2, 2**D** 2, 4

Classified Practice

Applications of Differentiation: Polynomial Functions

Q055 MCQ

The derivative sign table is used to determine:**Choose the correct option.**

- A only the y-intercept
- B increase/decrease and extrema
- C the constant term
- D the domain

Q056 MCQ**On a closed interval, an endpoint can be:****Choose the correct option.**

- A ignored
- B an absolute extreme
- C a derivative root only
- D never a minimum

Classified Practice

Applications of Differentiation: Polynomial Functions

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$

B $x = -\sqrt{2}$

C $x = \sqrt{2}$

D $x = 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

Q060 MCQ**In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:****Choose the correct option.**

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:**Choose the correct option.**

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

- A non-negative for all x
- B negative for all x
- C linear
- D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:**Choose the correct option.**A $[1, \infty)$ B $[-1, 1]$ C $(-\infty, 0)$ D $[0, 1]$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q065 MCQ

A cubic with both stationary values above the x-axis has:**Choose the correct option.**

- A three roots
- B one distinct real root
- C no real roots
- D two repeated roots

Q066 MCQ**To count cubic roots, compare the extrema with:****Choose the correct option.**

- A the y-axis
- B the x-axis
- C the derivative axis
- D the domain only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:**Choose the correct option.**

- A one distinct intersection
- B two distinct intersections
- C three distinct intersections
- D no intersection

Q068 MCQ**For $x^4 - 4x^3 + k$, the global minimum occurs at:****Choose the correct option.****A** $x = 0$ **B** $x = 1$ **C** $x = 3$ **D** $x = 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 3 Concept Quiz · Written**Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.**

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)**At a local minimum, f' changes:****Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Quiz MCQ 6 Concept Quiz · MCQ

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

- A always increasing
- B always decreasing
- C constant
- D undefined

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 Concept Quiz · MCQ

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$

B $k + 27$

C $k - 3$

D $27 - k$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4}$ or $\frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Classified Practice

Integration: Definite Integrals and Applications

Q001 **Written****Find the two indefinite integrals in the Warm Up.**

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Classified Practice

Integration: Definite Integrals and Applications

Q003 **Written****Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).**

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Classified Practice

Integration: Definite Integrals and Applications

Q005 Written

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 Written**Find the two given** Try functions from their derivatives and one given value of F .

Classified Practice

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.**

Q008 Written

Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.

Classified Practice

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.

Classified Practice

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Classified Practice

Integration: Definite Integrals and Applications

Q013 Written

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Classified Practice

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 Written**Solve the four given Try definite-integral transformations.**

Classified Practice

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.**

Q018 Written**Solve the final six given Exercise questions using the factorized definite-integral identity.**

Classified Practice

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Q020 Written**Solve the two given Try integral equations for $f(x)$.**

Classified Practice

Integration: Definite Integrals and Applications

Q021 Written

Find $f(x)$ for all five given integral equations.

Q022 Written**Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 Written**Differentiate the three given expressions with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q025 Written

Find $f(x)$ and a for the three given equations with a variable upper limit.

Q026 MCQ**The constant of integration is required because:****Choose the correct option.**

- A every primitive is unique
- B primitives differ by a constant
- C the integrand is constant
- D the limits are equal

Classified Practice

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Choose the correct option.**

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ**Before integrating $(x - 1)^2$, first:****Choose the correct option.**

- A differentiate it
- B expand it
- C set $x = 0$
- D add limits

Classified Practice

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Choose the correct option.**

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ**To find $F(x)$ from $F'(x)$, you should first:****Choose the correct option.**

- A differentiate again
- B integrate $F'(x)$
- C set $F = 0$
- D ignore the initial value

Classified Practice

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Choose the correct option.**

- A the derivative
- B the domain
- C the constant C
- D the leading coefficient

Q032 MCQ**If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:****Choose the correct option.**

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

Classified Practice

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Choose the correct option.**A $f(x)$ B $f'(x)$ C $f''(x)$ D $x + y$

Q034 MCQ**A point (x_0, y_0) on $y = f(x)$ gives:****Choose the correct option.**

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

Classified Practice

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

Classified Practice

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1

B a

C 0

D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:**Choose the correct option.**

- A $\int_a^b f = \int_b^a f$
 - B $\int_a^b f = -\int_b^a f$
 - C both are zero
 - D only if $f = 0$
-
-
-

Classified Practice

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Choose the correct option.**

- A integrate each factor separately
- B expand the product
- C differentiate the product
- D change the limits

Q040 MCQ**Adjacent integrals with the same integrand can be:****Choose the correct option.**

- A multiplied
- B combined over the joined interval
- C set equal to zero
- D reversed only

Classified Practice

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Choose the correct option.

- A different roots
- B the displayed roots (a,b)
- C zero coefficients
- D equal limits

Q042 MCQ

The identity for the factored integral is:**Choose the correct option.**

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:**Choose the correct option.**

- A squaring the integrand
- B reversing its limits
- C adding C
- D differentiating

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

- A a function of x
 - B a constant
 - C always zero
 - D a derivative
-
-
-

Classified Practice

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Choose the correct option.**

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A $1/3$

B $-1/3$

C 0

D 1

Classified Practice

Integration: Definite Integrals and Applications

Q047 MCQ

 $\frac{d}{dx} \int_a^x f(t)dt$ equals:

Choose the correct option.

A $f(a)$ B $f(x)$ C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:**Choose the correct option.**

- A the lower limit
- B the upper-limit variable
- C C
- D zero

Classified Practice

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

- A the derivative
- B the unknown lower limit a
- C the integrand degree
- D the upper limit

Q050 MCQ**For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:****Choose the correct option.**

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written**Evaluate** $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 5 **Concept Quiz · MCQ****The derivative of $\int_0^x (3t^2 + 1)dt$ is:****Choose the correct option.**

A $3x^2 + 1$

B $x^3 + x$

C $6x$

D 0

Quiz MCQ 6 Concept Quiz · MCQ

A primitive of $5x^4$ is:

Choose the correct option.

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Choose the correct option.

- A $\int_1^5 f(x)dx$
- B $\int_1^3 f(x)dx$
- C $\int_3^5 f(x)dx$
- D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.	Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.	Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.	Q036 B
Q002 Expand products or powers, collect like powers, then integrate every term.	Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.	Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.	Q037 C
Q003 Apply linearity and the power rule; retain the constant of integration.	Q014 Use linearity, expand products, then apply the endpoint-substitution rule.	Q022 $2x^2 + 3x - 5$.	Q038 B
Q004 Expand each product or square, simplify, and integrate term by term.	Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.	Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a .	Q039 B
Q005 $F(x) = x^3 + 4x^2 - x + 3$.	Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.	Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.	Q040 B
Q006 Integrate the derivative and use the given point to determine C .	Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.	Q025 Differentiate to get $f'(x)$, then set the integral to zero at $x = a$ to solve for a .	Q041 B
Q007 For each item, integrate $F'(x)$, then impose the stated value of F .	Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$.	Q026 B	Q042 B
Q008 $f(x) = x^3 - x^2 + x - 19$.	Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.	Q027 B	Q043 B
Q009 Integrate the given gradient and use the point on the curve to determine the constant.		Q028 B	Q044 B
Q010 Integrate the polynomial gradient and apply the given point condition in each case.		Q029 A	Q045 B
Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.		Q030 B	Q046 B
		Q031 C	Q047 B
		Q032 A	Q048 B
		Q033 B	Q049 B
		Q034 B	Q050 A
		Q035 B	Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
			Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
			Quiz WRITTEN 3 6
			Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
			Quiz MCQ 5 A
			Quiz MCQ 6 A
			Quiz MCQ 7 A
			Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

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Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Classified Practice

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.

Classified Practice

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.

Q004 Written**Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 Written**Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**

Classified Practice

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written**Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**

Classified Practice

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Q010 Written**Find the area bounded by $y = -x^3 + 4x$ and the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Classified Practice

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.

Q014 Written**Evaluate** $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 Written**Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.**

Classified Practice

Applications of Integration to Geometry

Q017 **Written**

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Classified Practice

Applications of Integration to Geometry

Q019 **Written**

Solve the **given** bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 MCQ**When the graph is below the x-axis, area is:****Choose the correct option.**

- A the signed integral
- B the negative of the integral
- C always zero
- D undefined

Classified Practice

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Choose the correct option.**

- A the graph and sign
- B only the y-intercept
- C the font size
- D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:**Choose the correct option.**A $8/3$ B $16/3$ C $32/3$ D $64/3$

Classified Practice

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Choose the correct option.**

- A lower-upper
- B upper-lower
- C product
- D derivative

Q024 MCQ**If curves cross inside the interval, area must be:****Choose the correct option.**

- A split at the crossing
- B ignored
- C doubled always
- D negative

Classified Practice

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Q026 MCQ**For cubic area, the sign changes at:****Choose the correct option.**

- A every turning point
- B each x-intercept
- C the y-intercept only
- D the midpoint

Classified Practice

Applications of Integration to Geometry

Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Choose the correct option.**

A 4

B 6

C 8

D 12

Q028 MCQ**A negative lobe contributes:****Choose the correct option.**

- A negative area
- B positive absolute area
- C no area
- D a derivative

Classified Practice

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Choose the correct option.**

- A the roots
- B the vertex only
- C the y-axis
- D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Choose the correct option.**A at c

B at 0 only

C never

D after doubling

Q032 MCQ**A tangent line at $x=a$ has slope:****Choose the correct option.**A $f(a)$ B $f'(a)$ C a^2

D the area

Classified Practice

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Choose the correct option.**

- A intersection x-coordinate
- B y-intercept only
- C vertex only
- D origin

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Choose the correct option.**A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 3

Classified Practice

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Choose the correct option.**

- A the same area
- B the same slope
- C the same root
- D zero area

Q036 MCQ**For $y=2x-x^2$, the total area above the x-axis is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Choose the correct option.**

- A area ratio
- B font ratio
- C root count
- D period

Q038 MCQ**In a bisection problem, the cut-off area is:****Choose the correct option.**

- A the total area
- B half the total area
- C twice the total area
- D zero

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 1 Concept Quiz · MCQ

The area under $y = -x^2 + 4$ and above the x -axis is:

Choose the correct option.

A $16/3$

B 8

C $32/3$

D $64/3$

Quiz MCQ 2 Concept Quiz · MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 3 **Concept Quiz · MCQ****Evaluate** $\int_{-2}^1 |x + 1| dx$:**Choose the correct option.**

- A 2
- B $5/2$
- C 3
- D $7/2$

Quiz MCQ 4 Concept Quiz · MCQ

The area enclosed by $y = x^2$ and tangents through (3, 8) is:

Choose the correct option.

A $1/3$

B $1/2$

C $2/3$

D $4/3$

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$ Q003 $\frac{32}{3}$ Q004 $\frac{32}{3}$ Q005 $\frac{32}{3}$ Q006 $\frac{64}{3}$ Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$ Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$ Q016 $\frac{64}{3}$ Q017 $\frac{2}{3}$ Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$ Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$

Concept 1 | Fundamentals of Random Variables

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Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Classified Practice

Fundamentals of Random Variables

Q001 **Written**

For the sum of two dice, construct the probability distribution and find $P(5 \leq X \leq 8)$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Classified Practice

Fundamentals of Random Variables

Q003 **Written**

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Classified Practice

Fundamentals of Random Variables

Q005 **Written**

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 Written**Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).**

Classified Practice

Fundamentals of Random Variables

Q007 Written

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Classified Practice

Fundamentals of Random Variables

Q009 **Written****Solve both given Exercise groups of transformed expectations for coins and for balls.**

Q010 Written

For black balls drawn from the stated box, find EX and VX .

Classified Practice

Fundamentals of Random Variables

Q011 Written

Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).

Q012 Written**Find EX and VX for all five given Exercise random variables.**

Classified Practice

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

Classified Practice

Fundamentals of Random Variables

Q015 **Written****Find the standard deviation for both given probability-table exercises.**

Q016 Written

For the number of winning tickets in the given draw, find VY and σY for $Y=2X+1$ and $Y=-3X-2$.

Classified Practice

Fundamentals of Random Variables

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Classified Practice

Fundamentals of Random Variables

Q019 Written

Solve the two given transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Q020 Written**Solve the two given Try problems involving net winnings and determining a,b for a die transformation.**

Classified Practice

Fundamentals of Random Variables

Q021 Written

Solve all four **given** transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 MCQ**A probability distribution must have probabilities that sum to:****Choose the correct option.****A** 0**B** 1**C** the number of outcomes**D** the mean

Classified Practice

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Choose the correct option.**A $1/6$ B $1/36$ C $2/36$

D 0

Q024 MCQ**The number of heads in three coins can be:****Choose the correct option.****A** 1,2,3**B** 0,1,2,3**C** 0,3 only**D** any integer

Classified Practice

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Choose the correct option.**

- A $\sum PX = x$
- B $\sum xPX = x$
- C largest x
- D variance squared

Q026 MCQ**The mean number of tails in four fair coins is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Choose the correct option.**

A 3

B 3.5

C 4

D 6

Q028 MCQ **$E(aX+b)$ equals:****Choose the correct option.****A** aEX **B** $aEX+b$ **C** $EX+b$ only**D** $a+b$

Classified Practice

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Choose the correct option.**A $E[X]^2$ B equal to EX

C a weighted square sum

D always zero

Q030 MCQ**If $EX=2$, then $E(3X+2)$ is:****Choose the correct option.****A** 6**B** 8**C** 10**D** 12

Classified Practice

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Choose the correct option.**

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D $\sqrt{V}$

Q032 MCQ**Variance can never be:****Choose the correct option.**

- A positive
- B zero
- C negative
- D fractional

Classified Practice

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Choose the correct option.**

- A draw the distribution
- B take a square root
- C ignore probabilities
- D add the values

Q034 MCQ

Standard deviation is:**Choose the correct option.**

- A VX
- B $\sqrt{V(X)}$
- C VX squared
- D EX

Classified Practice

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Choose the correct option.**

A squared units

B the same units as X

C no units

D units cubed

Q036 MCQ **$V(aX+b)$ equals:****Choose the correct option.****A** $aVX+b$ **B** a squared times VX **C** $|a|VX$ **D** $VX+b$

Classified Practice

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Choose the correct option.**A $a \sigma X$ B $|a| \sigma X$ C $a^2 \sigma X$ D $\sigma X+b$

Q038 MCQ**For $Y=X+3$, the variance is:****Choose the correct option.****A** $VX+3$ **B** VX **C** $3VX$ **D** $9VX$

Classified Practice

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Choose the correct option.**

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ**If $a > 0$ and $\sigma(aX+b)=3$, then a is:****Choose the correct option.****A $3\sigma X$** **B $3/\sigma X$** **C $\sigma X/3$** **D $3+b$**

Classified Practice

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Choose the correct option.**

- A variance only
- B standard deviation only
- C the mean only
- D nothing

Q042 MCQ**If $EX=2$ and $EX^2=7$, then VX is:****Choose the correct option.****A 3****B 5****C 7****D 9**

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Classified Practice

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz](#) · MCQ

If $VX=5$, $V(3X+1)$ is:

Choose the correct option.

A 5

B 8

C 15

D 45

Quiz MCQ 6 Concept Quiz · MCQ**For a fair die EX is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 4

Classified Practice

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz](#) · MCQ

A probability table total must be:

Choose the correct option.

A 0

B 1

C 2

D the mean

Quiz MCQ 8 Concept Quiz · MCQ

If $\sigma_X=4$, $\sigma(X-9)$ is:

Choose the correct option.

A 0

B 4

C 13

D 16

Answer key

Use the question number shown on each page.

- | | | | |
|---|---|---|---|
| Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$. | Q009 For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table. | Q019 Use $E(aX + b) = aEX + b$ and $\sigma(aX + b) = a \sigma X$, with $a > 0$. | Q034 B |
| Q002 Use the sample space or combination to list every possible X and sum the relevant probabilities. | Q010 $EX = 6/7$; $VX = 20/49$. | Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules. | Q035 B |
| Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values. | Q011 Construct each probability table, then use $VX = EX^2 - EX^2$. | Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule. | Q036 B |
| Q004 For four fair coins $EX = 2$; for the hypergeometric draw $EX = 4/3$. | Q012 Use the distribution table for each experiment and the variance identity. | Q022 B | Q037 B |
| Q005 Use $EX = \sum xPX = x$ after constructing each distribution. | Q013 The table gives $VX = 24/49$, so $\sigma X = 2 \sqrt{6/7}$. | Q023 B | Q038 B |
| Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice. | Q014 Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table. | Q024 B | Q039 B |
| Q007 Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table. | Q015 Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$. | Q025 B | Q040 B |
| Q008 Apply linearity for $aX + b$ and use the weighted square sum for X^2 . | Q016 Use $V(aX + b) = a^2 VX$ and $\sigma(aX + b) = a \sigma X$. | Q026 B | Q041 C |
| | Q017 Apply the squared multiplier to VX and the absolute multiplier to σX . | Q027 B | Q042 A |
| | Q018 Use the variance and standard-deviation transformation formulas for each Y. | Q028 B | Quiz WRITTEN 1 $EX = 3/2$ and $VX = 3/4$ |
| | | Q029 C | Quiz WRITTEN 2 $VY = 32$ and $\sigma Y = 4 \sqrt{2}$ |
| | | Q030 C | Quiz WRITTEN 3 12 |
| | | Q031 B | Quiz WRITTEN 4 10 |
| | | Q032 C | Quiz MCQ 5 D |
| | | Q033 A | Quiz MCQ 6 C |
| | | | Quiz MCQ 7 B |
| | | | Quiz MCQ 8 B |

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Classified Practice

Expected Value and Variance of Multiple Variables

Q001 **Written**

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q003 **Written**

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

Classified Practice

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Classified Practice

Expected Value and Variance of Multiple Variables

Q007 **Written****Two 500-point coins and two 100-point coins are tossed. Find EZ.**

Q008 Written**A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .**

Classified Practice

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 Written

In two independent raffles, buy two tickets from a $\frac{2}{10}$ winner raffle and two from a $\frac{1}{6}$ winner raffle. Find the expected prizes.

Classified Practice

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at $1, 2, 3$ and Y uniform on $2, 4, 6$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 Written**Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Q022 Written**Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.**

Classified Practice

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Q025 Written

If X , Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Q026 MCQ **$E(X+Y)$ equals:****Choose the correct option.****A** $EX-EY$ **B** $EX+EY$ **C** $EXEY$ **D** EX/EY

Classified Practice

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Choose the correct option.**

- A $EXEYEZ$
- B $EX+EY+EZ$
- C $EX-EY-EZ$
- D 0

Q028 MCQ**The expected heads in one fair coin toss is:****Choose the correct option.****A** 0**B** $1/2$ **C** 1**D** 2

Classified Practice

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Choose the correct option.**

A 0.5

B 1

C 2

D 4

Q030 MCQ**If $EX=2$ and $EY=5$, $E(X+Y)$ is:****Choose the correct option.****A** 3**B** 7**C** 10**D** 25

Classified Practice

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Choose the correct option.**A $aEX+bEY$ B $abE(XY)$ C $EX+EY$ D $a+b$

Q032 MCQ**The expected score of one fair 100-point coin is:****Choose the correct option.**

- A 25
- B 50
- C 100
- D 200

Classified Practice

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Choose the correct option.**

- A 14
- B 21
- C 28
- D 35

Q034 MCQ**A fixed entry fee changes:****Choose the correct option.**

- A variance only
- B the mean only
- C neither mean nor variance
- D standard deviation only

Classified Practice

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:**Choose the correct option.**A $1/5$ B $2/5$ C $1/2$ D $4/5$

Q036 MCQ **$E(XY) = EXEY$ requires:****Choose the correct option.**

- A $X = Y$
- B independence
- C zero means
- D equal variances

Classified Practice

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Choose the correct option.**A $1/2$

B 1

C $3/2$

D 3

Q038 MCQ**For a fair die, EY is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Choose the correct option.**

- A dependent
- B independent
- C identical outcomes
- D impossible

Q040 MCQ**If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:****Choose the correct option.****A** 5**B** 6**C** 1**D** 9

Classified Practice

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X,Y, $V(X+Y)$ equals:**Choose the correct option.**A $VX-VY$ B $VX+VY$ C $VXVY$

D 0

Q042 MCQ

The variance of a fair die is:**Choose the correct option.**A $7/2$ B $35/12$ C $35/6$ D $1/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Choose the correct option.**A $V(X+Y)$ B $\sqrt{V(X+Y)}$ C $VX+VY$ D $E(X+Y)$

Q044 MCQ**The variance of the sum of two independent fair dice is:****Choose the correct option.**A $35/12$ B $35/6$ C $70/6$ D $49/4$

Classified Practice

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Choose the correct option.**

- A variance
- B standard deviation
- C neither spread measure
- D both spread measures

Q046 MCQ

For independent X, Y, Z , $E(XYZ)$ equals:**Choose the correct option.**A $EX+EY+EZ$ B $EXEYEZ$

C 0

D $E(XY)+EZ$

Classified Practice

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Choose the correct option.**A $VXVYVZ$ B $VX+VY+VZ$ C $VX-VY-VZ$

D 0

Q048 MCQ**For a binomial count with $n=5$ and $p=1/6$, EX is:****Choose the correct option.****A** $1/6$ **B** $5/6$ **C** $5/36$ **D** 6

Classified Practice

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Choose the correct option.**

A 2

B 3

C 4

D 6

Q050 MCQ**For three independent fair dice, $V(\text{sum})$ is:****Choose the correct option.**A $35/12$ B $35/6$ C $35/4$ D $343/8$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 Concept Quiz · Written

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 Concept Quiz · MCQ

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Choose the correct option.

- A 7
- B 10
- C 25
- D $1/2$

Classified Practice

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For two independent die rolls, $V(X+Y)$ is:

Choose the correct option.

A $35/12$

B $35/6$

C $35/4$

D $7/2$

Quiz MCQ 8 Concept Quiz · MCQ

For three independent variables, $V(X+Y+Z)$ is:

Choose the correct option.

- A the product of variances
- B the sum of variances
- C the sum of means
- D always zero

Answer key

Use the question number shown on each page.

- Q001 $EX = 35/3, EY = 9/2, \text{so } E(X + Y) = 97/6.$ Q016 $VX=2/3, VY=3/4, \text{so } V(X + Y) = 17/12.$ Q025 $E(XYZ) = 23/21=3.$
- Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$ and $\sigma = \sqrt{17/12}.$ Q026 B
- Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$ Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$ Q027 B
- Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$ $\sigma = \sqrt{35/6}.$ Q028 B
- Q005 Each mean is $2(0.5) = 1,$ so Q018 $V = 1/4 + 35/12 = 19/6;$ Q029 B
- $E(X + Y + Z) = 3.$ $\sigma = \sqrt{19/6}.$ Q030 B
- Q006 $Z = 500X + 100Y, \text{so } EZ = 5001 + 1003/2 = 6501.5$ Q019 $V = 21/4 + 31/4 = 5/4.$ Q031 A
- Q007 $EZ = 5001 + 1001 = 600.$ Q020 $V = 5/4 + 35/12 = 25/6;$ Q032 B
- Q008 $EZ = 37/2 + 57/2 = 28.$ $\sigma = 5/\sqrt{6}.$ Q033 C
- Q009 $EZ = 350 + 25 = 175.$ Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and } V(sum) = 8.$ Q034 B
- Q010 $EZ = 22/10 + 21/6 = 11/15.$ Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and } V(sum) = 8.$ Q035 B
- Q011 X and Y are independent, so Q023 $E(product) = 7/2^3 = 343/8;$ Q036 B
- $E(XY) = EXEY = 3/27/2 = 21/4.$ $V(sum) = 335/12 = 35/4.$ Q037 C
- Q012 $EX = 8/5, EY = 4, \text{so } E(XY) = 32/5.$ Q024 Means are $5/6, 2/3, 1/2;$ Q038 C
- Q013 $E(XY) = EXEY = 3/21 = 3/2.$ $E(product) = 5/18 \text{ and } V(sum) = 5/3.$ Q039 B
- Q014 $EX = EY = 5/2, \text{so } E(XY) = 25/4.$ Q040 B
- Q015 $E(XY) = EXEY = 2/7.$ Q041 B
- Q042 B
- Q043 B
- Q044 B
- Q045 C
- Q046 B
- Q047 B
- Q048 B
- Q049 C
- Q050 C
- Quiz WRITTEN 1 -2
- Quiz WRITTEN 2 10
- Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$
- Quiz WRITTEN 4 $9/4$
- Quiz MCQ 5 B
- Quiz MCQ 6 B
- Quiz MCQ 7 B
- Quiz MCQ 8 B